

ASSESSMENT - PSEUDO CODE WRITING.
Q1: DESIGNING CLASS HIERARCHY (PSEUDO CODE)

CLASS Person

DATA

name

age

METHOD Person(name, age)

 this.name ← name

 this.age ← age

END METHOD

END CLASS

CLASS Student Inherits person

DATA

registerNumber

marks[5]

METHOD Student(name, age, registerNumber, marks[5])

 SUPER(name, age)

 this.registerNumber ← registerNumber

 this.marks ← marks

END METHOD

METHOD calculateAverage()

 total ← 0

 FOR i ← 0 TO 4

 total ← total + marks[i]

 END FOR

 RETURN total/5

END METHOD

END CLASS

```
CLASS ResearchStudent INHERITS Student
```

```
DATA
```

```
    researchArea.
```

```
METHOD ResearchStudent(name, age, registerNumber,  
                           marks[5], researchArea)
```

```
    SUPER(name, age, registerNumber, marks)
```

```
    this.researchArea ← researchArea
```

```
END METHOD
```

```
METHOD displayCompleteDetails()
```

```
    PRINT "Name: " + name
```

```
    PRINT "Age: " + age
```

```
    PRINT "Register Number: " + registerNumber
```

```
    PRINT "Average Marks: " + calculateAverage()
```

```
    PRINT "Research Area: " + researchArea
```

```
END METHOD
```

```
END CLASS
```

```
BEGIN
```

```
    r.s ← NEW ResearchStudent("Anitha", 24, "R1023",  
                               [88, 91, 79, 85, 93], "machine learning")
```

```
    r.s displayCompleteDetails()
```

```
END
```